

Selim Amrari Mariette



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Sr Data Scientist with 6 years of experience working at top-ranking startups. My key areas of specialization include Image processing, Multivariate time series forecasting, Text classification, Geospatial processing

Education

Master of Science **EPFL** September 2015 - February 2018
Environmental and mathematical modeling
"École Polytechnique fédérale de Lausanne"

Professional experience

Freelance - Sr Data Scientist May 2025 - Now

- Product: built a decision-support dashboard for visualizing statistics and predicting customer flow for restaurants.
[Multivariate time series forecasting, Transformers Network](#)

5-months career break to enjoy fatherhood - Apprentice Dad December 2024 - April 2025

Calyps/Saniia (spin-off) - Sr Data Scientist October 2021 - November 2024

- Product: Improved models for hospital emergency flow forecasting products, through feature engineering, critical bugs identification and external data sources ingestion. Decreased prediction error by about 15%.
[Multivariate time series forecasting, Transformers Network, Web Scraping](#)
- Customer relationship management: conducted weekly meeting with product users
- Created three PoCs for customers demand forecasting and designed MVPs
[Transformers Network](#)
- Designed PoCs focused on NLP tasks such as automation of billing classification using text classification
[Transformers Network, Hugging Face](#)
- Infrastructure: Built a containerized infrastructure for all the company products, reducing deployment failures and improving scalability. Mentored the team on Docker.
[Docker](#)
- Led technical interviews and designed assessments for data scientist positions

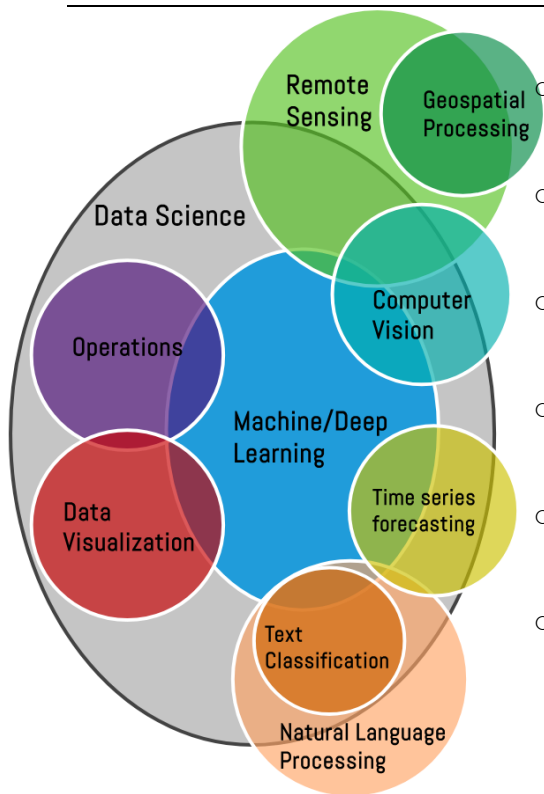
Gamaya - Sr Data Scientist February 2019 - September 2021

- Product: Developed a deep learning model, used by all the team, and combining recurrent and convolutional architecture dedicated to spatio-temporal analysis of satellite images
[Anomaly Detection, Sequential Neural Network](#)
- Infrastructure/Product: Built an AI product dedicated to large scale sugarcane crop monitoring, estimating growth and highlighting anomaly automatically, used by customers to monitor thousands of parcels.
[Sequential and Convolutional Neural Networks](#)
- Infrastructure: Implemented a drone/satellite images alignment (coregistration) tool as a preprocessing building block.
[Computer vision algorithms](#)
- Infrastructure: Engineered an automated cloud detection pipeline utilizing a deep learning model and processing 100% of incoming satellite imagery.
[Convolutional Neural Networks](#)
- PoC: Conducted tests of Computer vision algorithms for sowing lines detection.
[Image processing approaches: Dynamic thresholding, Delineation, Clustering](#)
- Mentored a Master-Intern on bio-physical parameters inversion project and supervised two products developed with subcontractors.

IFREMER - Research Engineer August 2017 - September 2018

- Built a satellite-based bathymetry retrieval model delivering regular data for lagoon hydrodynamic modeling on thousands squared kilometers of coastal areas.
[Robust Regressions, Probabilistic mixtures models](#)

Technical skills and programming tools



- **Broad expertise in M-L and Deep Learning:**

TensorFlow, Keras, PyTorch Lightning, Pymc3, Scikit, Statsmodel

- **Natural Language Processing:**

NLTK, Gensim, Huggingface

- **Image Processing and Computer Vision:**

OpenCV, Scikit-image, photogrammetry

- **Remote Sensing and Geospatial Processing:**

GDAL, OGR, Geoda, Geopandas, Qgis

- **Data handling and visualization:**

Pandas, Dask, Xarray, Bokeh, streamlit, holoviews, bokeh

- **Expertise in ML Ops:**

Docker, Git, DVC, GCP, MLflow, Web Scraping, bash, Linux



Additional Experience

Natural Language Processing project:

- ★ Automated papers assignation to journal using LLM and Text classification : [presentation and repo](#)
- ★ Leveraged latent space from pretrained LLM to achieve zero-shot classification : [presentation and repo](#)

Multivariate time series forecasting projects:

- ★ Electricity demand forecasting: [presentation and demo](#)
- ★ Designed decision-support dashboard for restaurant customer flow forecasting: [presentation and demo](#)
- ★ Built Auto-Updating COVID-19 Predictions Web Dashboard App : [project page](#)

Computer Vision and Remote sensing projects:

- ★ Developed Satellite Derived Bathymetry model optimized for lagoon ecosystem: [presentation](#)
- ★ Classified of urban tree species based on hyperspectral and LIDAR data : [presentation and repo](#)
- ★ Achieved Road segmentation from aerial imagery based on convolutional neural networks : [report](#)
- ★ Designed a deep learning model able to ingest raw satellite time series for change detection: [repo](#)

Classical projects:

- ★ Customers churn detection

Language

French: Native language
English: Advanced
German: basic knowledge

Personal details

Happy young dad
Nationalities: Swiss, French
Birth: 08 november 1990 in Fribourg
Kitesurfer for 10 years, now surviving on wingfoil in Swiss Lakes
Definitely team ski – sorry, snowboarders
Ran a 40km trail once, hoping to catch up with my old shape...
still walking, waiting for it